Evaluating performance of an AI-based recurrence score in early breast cancer patients treated with chemoendocrine therapy: a secondary analysis of the UNIRAD trial

ATARAXIS

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Introduction

- Patients with node-positive HR+/HER2- early breast cancer are at high risk for relapse within 5 years of diagnosis, underscoring the need for treatment escalation.[1,2]
- However, the UNIRAD trial did not demonstrate an overall benefit from adding everolimus to adjuvant endocrine therapy in this population.[3]
- Precision medicine tests have provided physicians with prognostic information beyond traditional staging criteria, thereby allowing more individualized therapy decision-making in women with breast cancer.

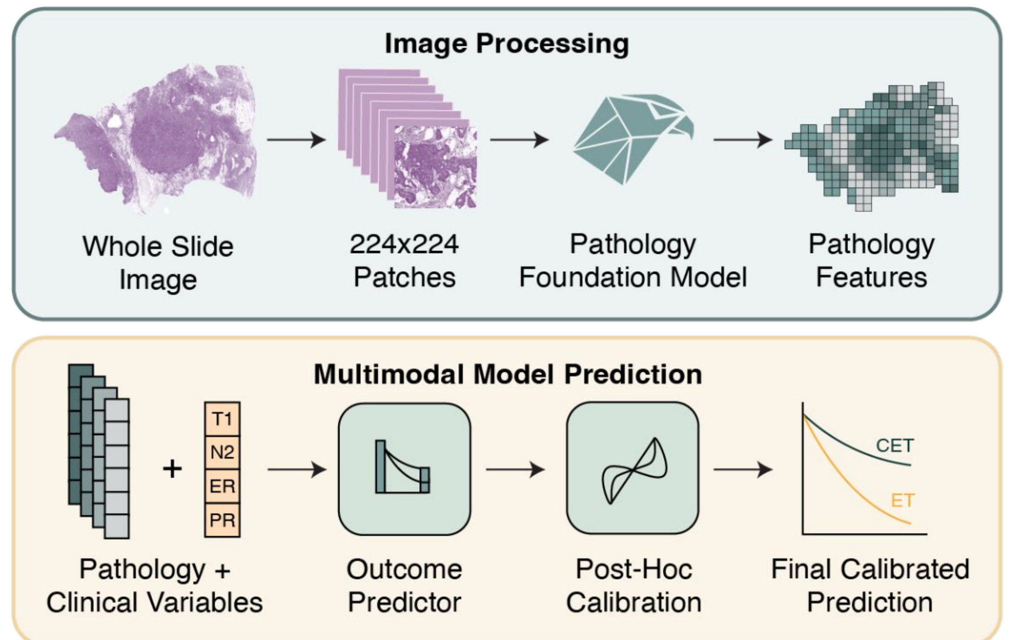

Figure 1: Schematic of ATX pipeline integrating pathology and clinical variables.

Ataraxis Breast (ATX)[4] is an artificial intelligence tool that integrates morphological features extracted from H&E-stained slides with clinical features to estimate recurrence risk.

- We evaluated ATX's ability to:
  1. Establish the RoR of ATX in patients with early stage HR+ HER2- breast cancer enrolled in the phase III UNIRAD clinical trial.
  2. Assess ATX-based RoR performance within UNIRAD trial treatment arms.

Methods

- ATX risk scores were generated using a locked model from 689 patients enrolled in the UNIRAD trial, including patients from both trial arms. No patients from the UNIRAD trial were used in the training of ATX.
- Patients were classified into high and low risk groups based on a pre-specified threshold.
- Kaplan-Meier estimators were used to predict the probability of meeting the DFS endpoint, and differences between ATX-defined low and high risk groups were assessed using log-rank tests.
- To quantify relative differences in the hazard of experiencing an event contributing to the disease-free survival endpoint associated with ATX scores, Cox proportional hazards models were fit, from which hazard ratios (HRs) were estimated.

Patients

- Of the 689 patients, 340 (49%) were classified as high risk and 349 (51%) were classified as low risk.
- Within the control arm, 165 patients (49%) were classified as high risk and 170 (51%) were classified as low risk.
- In the everolimus arm, 175/354 (49%) were classified as high risk.

Results

5-Year Recurrence Across All Patients

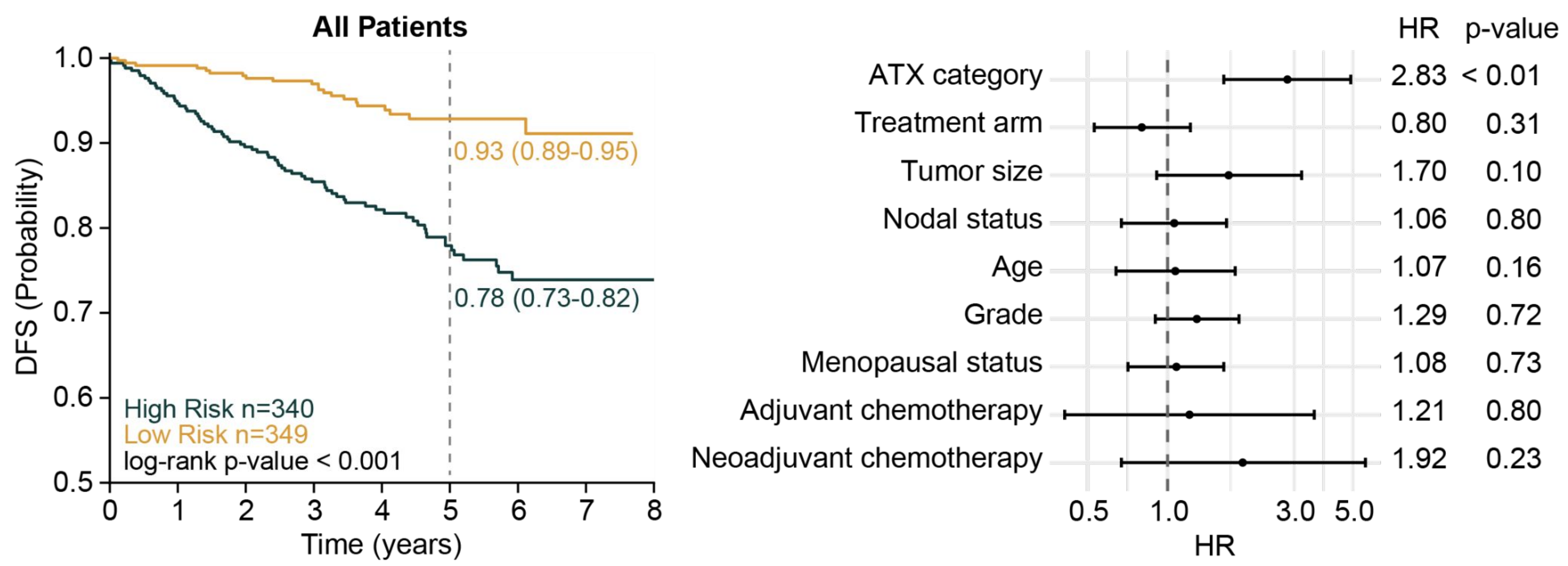

Figure 2: Across all patients, ATX performance stratifies disease-free survival.

- ATX high risk classification was significantly associated with worse DFS (HR = 2.83, p < 0.01, Figure 2).
- After adjusting for treatment arm and all clinical covariates, ATX score remained independently and significantly associated with DFS.
- ATX was the only variable significantly associated with DFS.

5-Year Recurrence Across Control Arm Patients

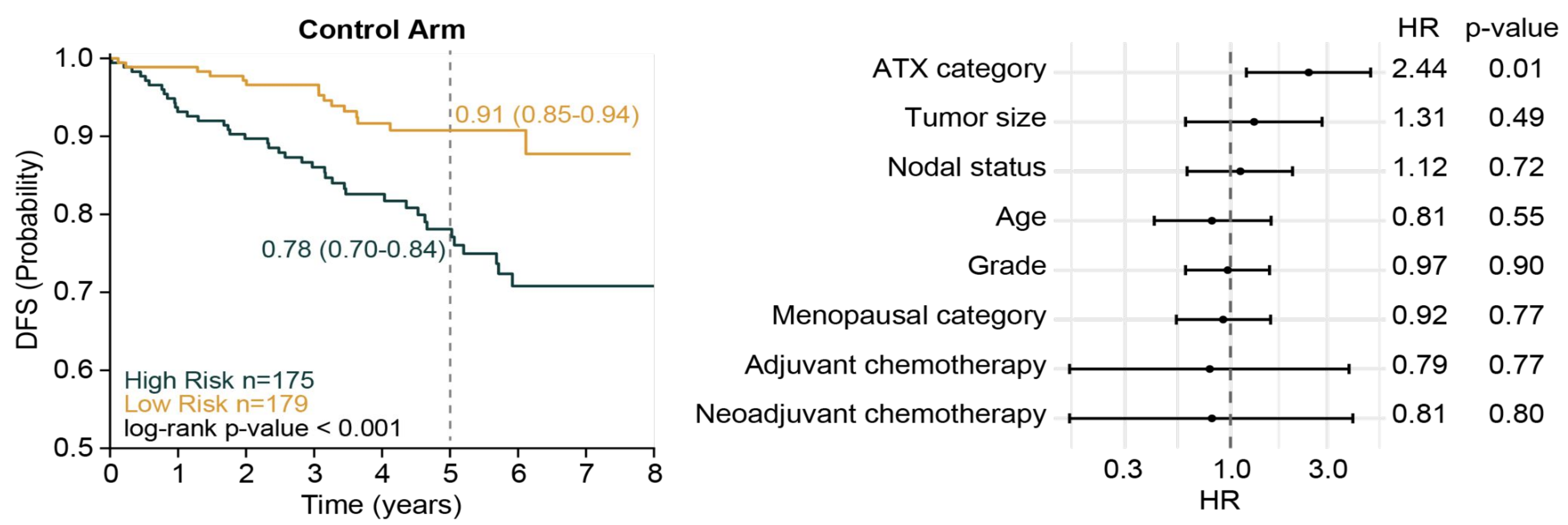

Figure 3: ATX performance stratifies disease-free survival across patients in the control arm.

- ATX retained strong prognostic performance in patients who were randomized to the control arm of the UNIRAD trial (HR = 2.44, p = 0.01, Figure 3).
- After controlling for tumor size, nodal status, age, grade, menopausal status, receipt of adjuvant chemotherapy, and receipt of neoadjuvant chemotherapy, ATX score remains prognostic.
- ATX remained the only significant variable after multivariable adjustment.

Results Continued

5-Year Recurrence Across Everolimus Arm Patients

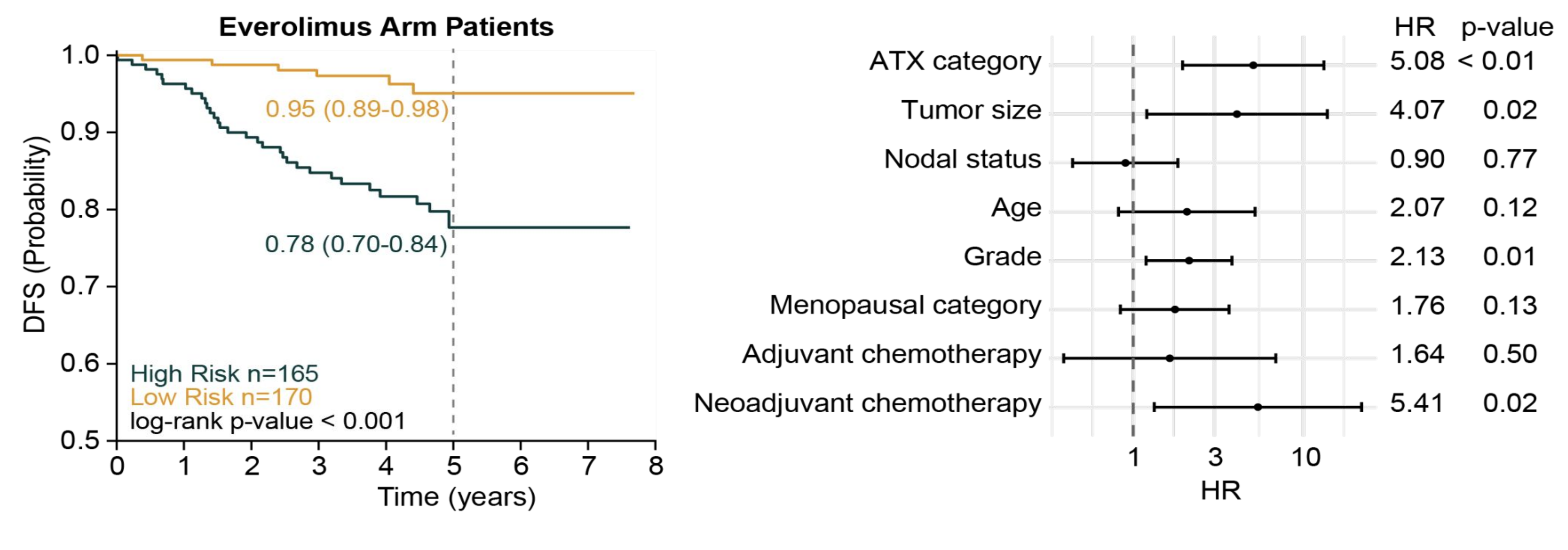

Figure 4: ATX performance stratifies disease-free survival across patients in the everolimus arm.

- ATX score was significantly prognostic in patients randomized to the everolimus arm (HR = 5.08, p < 0.01, Figure 4).
- After multivariate adjustment, ATX score remains prognostic.
- In this arm, tumor size, tumor grade, and receipt of neoadjuvant chemotherapy are also significantly associated with DFS.
- This pattern was not observed in the control arm.

Conclusions

- ATX high risk patients treated with standard-of-care therapy had significantly increased hazard of an DFS-contributing event in a clinically homogenous trial cohort of patients with node-positive HR+/HER2- early breast cancer.
- These findings suggest that AI-based risk stratification identifies biologically high-risk patients who may derive benefit from adjuvant treatment escalation.

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References

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